

EXECUTIVE DEMAND INTELLIGENCE BRIEFING

EXECUTIVE SUMMARY

This technical brief outlines the deployment of an enterprise-tier Demand Intelligence platform built using multi-component time-series models, unsupervised cluster allocations, and dual-layered anomaly detection grids. The platform has been fully containerized into a live interactive dashboard (app.py), bridging the gap between deep statistical modeling and day-to-day warehouse logistics.

By prioritizing mathematical parameters over intuition, the system successfully stabilizes inventory overhead, profiles sub-category market growth, and provides active warning parameters against unexpected transactional disruptions.

SECTION 1: TIME SERIES DECOMPOSITION & FORECASTING INSIGHTS

1. Structural Signal Analysis

Through additive time-series decomposition applied to 48 months of historical records, two primary structural behaviors were isolated:

- **Long-Term Underlying Trend:** Core business velocity has grown linearly. The baseline shifted from an average of roughly \$34,000 monthly in 2015 to more than \$55,000 by late 2018.
- **Dominant Annual Seasonality:** The enterprise experiences highly predictable cyclical waves. Demand bottoms out significantly every January and February before escalating toward a heavy consumer spike in Q4 (specifically November and December).

2. Predictive Performance Matrix (Task 3 Validation)

To identify the most accurate engine for production deployment, three distinct architectures were evaluated on a hidden 3-month validation horizon:

Forecasting Model	Mean Absolute Error (MAE)	Root Squared Error (RMSE)	Mean Error	Mean Percentage Error (MAPE)	Absolute Error

SARIMA (1,0,1)x(1,0,1) ₁₂	19,730.20	24,677.26	19.67%
Facebook Prophet	20,296.01	22,487.47	21.89%
XGBoost Regressor	29,364.30	29,366.84	32.78%

- Production Selection Justification: SARIMA is selected as our production-tier forecasting model due to its clear statistical dominance. Machine Learning architectures like XGBoost failed significantly during testing because tree-based algorithms struggle to naturally extrapolate macro-economic trends on compact, row-constrained time-series frames without manual trend removal.

3. Granular Growth Projections (Task 4 Segment Horizons)

Extrapolating our winning SARIMA parameters into the upcoming Q1 2019 horizon across categories and regions reveals key operational shifts:

- The Technology Category represents our most explosive growth vector, climbing 86.37% from its February floor (\$17,491.72) to reach \$32,599.60 by March.
- The West Region represents our leading geographic zone, displaying a massive 129.74% demand surge over that same March window (\$12,166.55 to \$27,951.79).

SECTION 2: OPERATIONAL RISK DEFENSE & INVENTORY STRATIFICATION

1. Automated Supply Chain Anomaly Auditing

To protect our supply chain from sudden bottlenecks, an unsupervised Isolation Forest density model was implemented. The system identified 9 global anomaly weeks across the historical timeline, highlighting distinct operational states:

- Sustained Q4 Volume Surges (e.g., Late November): Represent high-revenue, high-volume promotional anomalies (Black Friday/Cyber Monday). Logistics teams must handle these as planned surges by securing short-term warehouse staff and shipping capacity well in advance.
- Year-End Drops (Late December): Represent a standard post-holiday freeze. Operations should use this predictable dip to execute warehouse audits and stock counts without interrupting core fulfillment.
- Note: The local rolling Z-Score returned 0 flags due to the rapid, sustained nature of our Q4 growth blocks, which expanded variance thresholds. This proves that

an absolute density framework like Isolation Forest is essential for catching multi-week demand shifts.

2. K-Means Clustering Stocking Recommendations

Using the K-Means Elbow Method ($k=4$), product sub-categories were divided into clear inventory tracking groups to replace uniform stocking practices:

- Cluster 0 — High Value, Low Volatility Outliers (['Copiers']): Implement a Just-In-Time (JIT) Procurement workflow. Do not freeze capital holding massive quantities of these high-cost assets in standard storage cells.
- Cluster 1 — Low Volume, Low Volatility (['Appliances', 'Art', ...]): Establish automated Fixed Reorder Points (ROP) inside ERP systems to automatically retrigger bulk baseline replenishment requests when minimum safety buffers are crossed.
- Cluster 2 — Growing Demand, Medium Volatility (['Accessories', 'Binders', ...]): Implement an Aggressive Safety Stock Buffer. Increase active baseline shelf counts by 15-20% to seamlessly capture rapid growth vectors without risking stockouts.
- Cluster 3 — High Volume, High Volatility (['Chairs', 'Machines', ...]): Enforce Dynamic Demand Alignment. Synchronize monthly rolling warehouse intake directly with our live SARIMA forecasting updates to balance storage overhead costs against shifting demand.

SECTION 3: REPOSITORY COMPLETION CHECKLIST

Verify that the workspace directory contains exactly the following file layout for final archiving:

Plaintext

```
Week3&4/
|
├── analysis.ipynb      # Complete notebook containing all 6 task code blocks &
markdown cells
├── train.csv          # Superstore sales master data file
├── app.py             # The production Streamlit application code block
├── requirements.txt    # Frozen pip library dependencies
├── summary.pdf        # The 2-page executive report outlined above
|
└── charts/            # Extracted visual outputs
    ├── time_series_decomposition.png
    ├── prophet_components.png
    ├── actual_vs_predicted.png
    ├── segmented_forecasts.png
    ├── anomaly_detection_map.png
    └── clustering_elbow_curve.png
```

└─  product_demand_clusters.png

Application Deployment Context: Live Production Deployment Verified via Streamlit Cloud

App URL Link:

<https://sales-forecasting-demand-intelligence-nvkpkku4xgcgbjeyhqdur6.streamlit.app/>